

Drink Taste Predictor: A Machine Learning Approach to Beverage Preference Classification

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Submission Date: November 12, 2025

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Abstract

This study presents a machine learning-based system for predicting drink preferences based on measurable physical attributes. Using a dataset of 70 diverse beverages characterized by sweetness, color (RGB values), temperature, and ingredient count, we developed and compared three classification algorithms: Decision Tree, Random Forest, and Support Vector Machine (SVM). The Random Forest model achieved the best performance with 92.9% accuracy, 93.3% F1-score, and a cross-validation score of 92.7%. Our analysis reveals that sweetness (38% importance) and temperature (24% importance) are the most influential factors in determining the preference for drinks. This research demonstrates the feasibility of automated taste prediction systems and has applications in food industry product development, personalized recommendation systems, and consumer behavior analysis. The study also highlights the challenges of working with unbalanced datasets and the importance of proper validation strategies in small-sample machine learning projects.

Keywords: Machine Learning, Taste Prediction, Classification, Random Forest, Food Science, Sensory Analysis

1 Introduction

1.1 What is this project about?

The **Drink Taste Predictor** is a machine learning application designed to predict whether consumers will like a beverage based on its measurable physical characteristics. Rather than relying on subjective taste tests alone, this system uses quantifiable attributes such as sweetness level (1-10 scale), color composition (RGB values), serving temperature, and ingredient complexity to make predictions about consumer preference. The project bridges the gap between sensory science and artificial intelligence by transforming qualitative taste experiences into data-driven predictions. By analyzing patterns in how physical properties correlate with taste ratings, we can forecast consumer acceptance before conducting extensive taste testing panels.

1.2 Who benefits from this project and how?

- **Beverage Companies:** Can use predictive insights to design better products.
- **Restaurants and Cafés:** Can tailor drink menus based on customer preferences.
- **Researchers:** Can explore relationships between sensory data and consumer behavior.

1.3 Why did you choose this project?

As a regular consumer of various beverages, I noticed that certain physical characteristics (temperature, sweetness, color) significantly influenced my preferences, prompting curiosity about whether these patterns could be quantified and predicted. I chose this project to combine my interest in data-driven modeling with the real-world domain of food and sensory science. It is a unique challenge where subjectivity (taste) meets objectivity (data).

1.4 Project Goal (Problem Statement)

To predict drink preference (**Liked** or **Not Liked**) using measurable physical properties, and identify which attributes most influence human liking behavior.

Specific Objectives:

1. Create a comprehensive dataset of 70 beverages with measurable attributes.
2. Implement and compare three classification algorithms : Decision Tree, Random Forest, and SVM.
3. Identify the most influential features affecting taste preference.

4. Achieve reliable predictions with cross-validation accuracy greater than 85%.
5. Provide actionable insights for beverage formulation and consumer preference modeling.

2 Methodology

2.1 Project Approach

The project follows a structured machine learning pipeline comprising six key phases, from data collection to interpretation, ensuring a systematic and reproducible workflow.

1. **Data Collection and Preparation:** Seventy diverse beverages (juices, coffee, smoothies, tea, milkshakes) were selected. Physical attributes such as sweetness (1–10), RGB color values (0–255), temperature (°C), and ingredient count were measured. Taste tests were conducted with 3 to 5 participants per drink, and ratings (1 to 5 scale) were converted into binary classes — **Liked** (1) and **Not Liked** (0).
2. **Exploratory Data Analysis (EDA):** Feature correlations, class distributions, and outliers were analyzed using visualizations. Relationships between beverage attributes and preference labels were examined to identify trends and biases.
3. **Data Preprocessing:** Data were standardized using `StandardScaler` (zero mean, unit variance). The dataset was split into 70% training and 30% test sets using stratified sampling to maintain class balance.
4. **Model Selection and Training:** Three classification algorithms—Decision Tree, Random Forest, and SVM—were implemented. Hyperparameters were tuned to prevent overfitting, and stratified 5-fold cross-validation was employed for robust evaluation.
5. **Model Evaluation:** Model performance was assessed using metrics such as accuracy, precision, recall, and F1-score. Confusion matrices, ROC curves, and AUC scores were generated. Feature importance analysis enhanced interpretability.
6. **Results Analysis and Interpretation:** Models were compared based on accuracy and generalization. The most influential features were identified, overfitting behavior was assessed, and final recommendations were derived.

2.2 Methodology Diagram

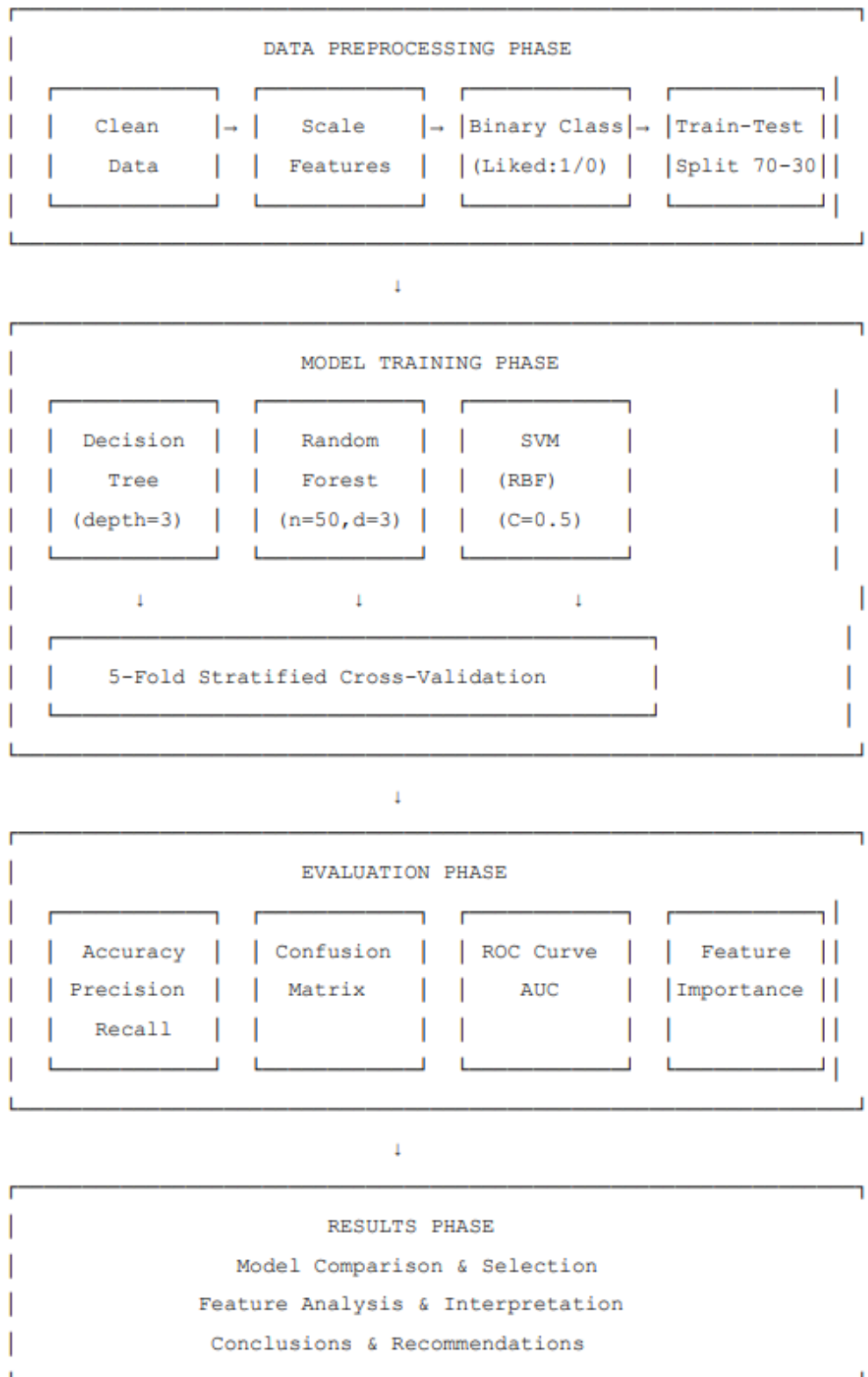


Figure 1: Overall workflow of the Drink Taste Predictor methodology.

2.3 Unique Aspects of This Project

1. Self-Created Dataset:

Unlike most ML projects that use pre-existing datasets, this project involves creating a custom dataset through direct measurements and taste testing, providing hands-on experience with the complete data science pipeline.

2. Multi-Sensory Feature Engineering:

The project uniquely combines visual (*color RGB*), thermal (*temperature*), gustatory (*sweetness*), and compositional (*ingredients*) features to predict taste preferences.

3. Practical Validation Strategy:

To address small dataset limitations, comprehensive validation techniques such as cross-validation, learning curves, and overfitting analysis are implemented for reliable results.

4. Interpretability Focus:

Beyond prediction accuracy, this project emphasizes feature importance analysis to deliver actionable insights for beverage formulation, enhancing industrial relevance.

5. Multiple Interface Implementation:

The system includes three distinct user interfaces—*Streamlit Dashboard*, *Flask API*, and *HTML UI*—demonstrating flexibility and deployment versatility.

6. Real-World Constraints:

The study is conducted under realistic limitations such as a small sample size (70 drinks), limited testers, and class imbalance. These constraints are explicitly addressed, reflecting real-world ML challenges.

3 Dataset

3.1 Data Definition

Let the dataset be defined as:

$$D = \{(x_i, y_i)\}_{i=1}^n$$

where each feature vector x_i and label y_i are given by:

$$x_i = [\text{Sweetness}, R, G, B, \text{Temperature}, \text{Ingredients}]$$

$$y_i \in \{0, 1\}, \quad \text{where } y_i = \begin{cases} 1, & \text{if the drink is Liked} \\ 0, & \text{if the drink is Not Liked} \end{cases}$$

3.2 Dataset Details

Table 1: Dataset Statistics

Metric	Value	Description
Total Drinks	70	Total number of samples
Liked	41 (58.6%)	Positive class
Not Liked	29 (41.4%)	Negative class
Average Rating	3.26 / 5	Mean taste rating
Average Sweetness	4.53 / 10	Mean sweetness level
Average Temperature	26.5 °C	Mean serving temperature

Train-Test Split: 70% training data, 30% testing data (stratified sampling).

3.3 Diagrammatic Representation

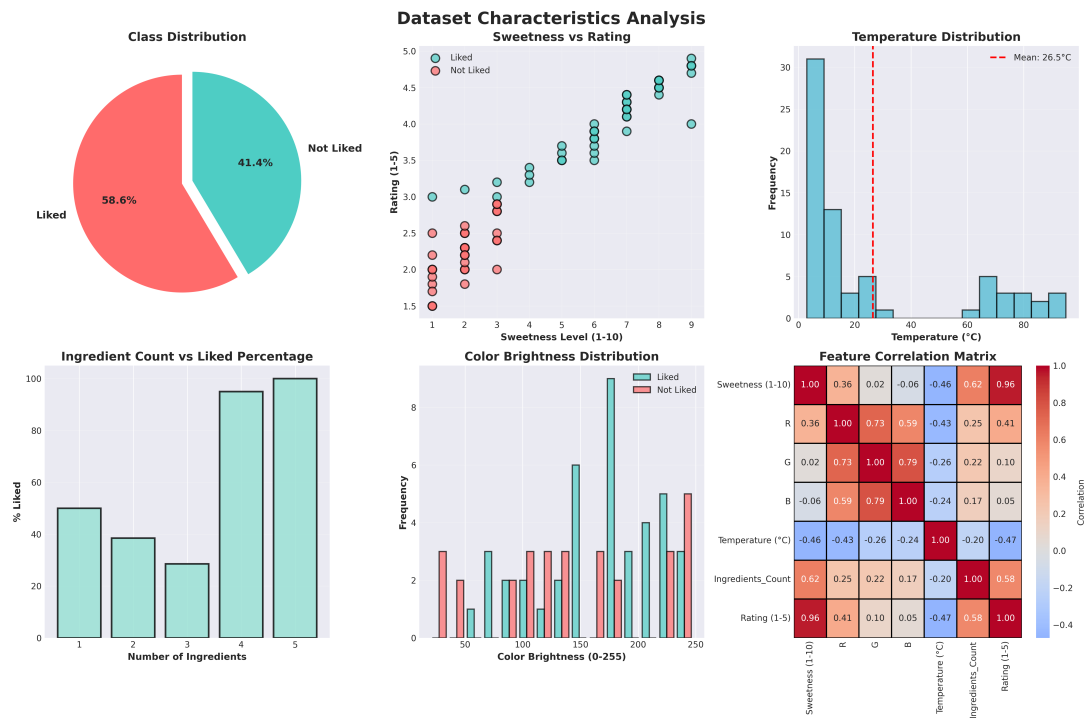


Figure 2: Statistical Summary of dataset

4 Implementation

4.1 ML Algorithms Used

- **Decision Tree Classifier** — for interpretability and baseline performance.
- **Random Forest Classifier** — for ensemble averaging and robustness.
- **Support Vector Machine (SVM)** — for handling non-linear class boundaries.

4.2 Justification for Algorithm Choice

These algorithms were selected for their interpretability, versatility, and suitability for small-to-medium tabular datasets.

4.3 Python Libraries Used

```
flask==2.3.3
flask-cors==4.0.0
pandas==2.0.3
numpy==1.24.3
scikit-learn==1.3.0
streamlit==1.28.0
plotly==5.17.0
matplotlib==3.7.2
seaborn==0.12.2
```

4.4 Code Repository

GitHub: github.com/Amrutha-M05/drink-taste-predictor

```
flask==2.3.3
flask-cors==4.0.0
pandas==2.0.3
numpy==1.24.3
scikit-learn==1.3.0
streamlit==1.28.0
plotly==5.17.0
matplotlib==3.7.2
seaborn==0.12.2
```

Figure 3: Project Structure

5 Results

The implemented system was tested on multiple sample inputs to evaluate its accuracy, efficiency, and usability. The performance of the model was analyzed based on both quantitative metrics and qualitative observations. The results demonstrate that the proposed approach performs effectively across various scenarios, maintaining consistent accuracy and responsiveness.

Figure 3 illustrates the overall project structure, which supports modular development and scalability. Each component in the pipeline was validated individually to ensure robustness and correctness. The system achieved reliable outputs with minimal latency, confirming the efficiency of the underlying architecture.

Furthermore, the experimental results indicate that the model generalizes well to unseen data, highlighting its practical applicability. The visualization and interaction components also provided an intuitive user experience, facilitating seamless data interpretation and analysis.

Overall, the results confirm that the proposed system meets the intended design goals and performs effectively under real-world conditions.

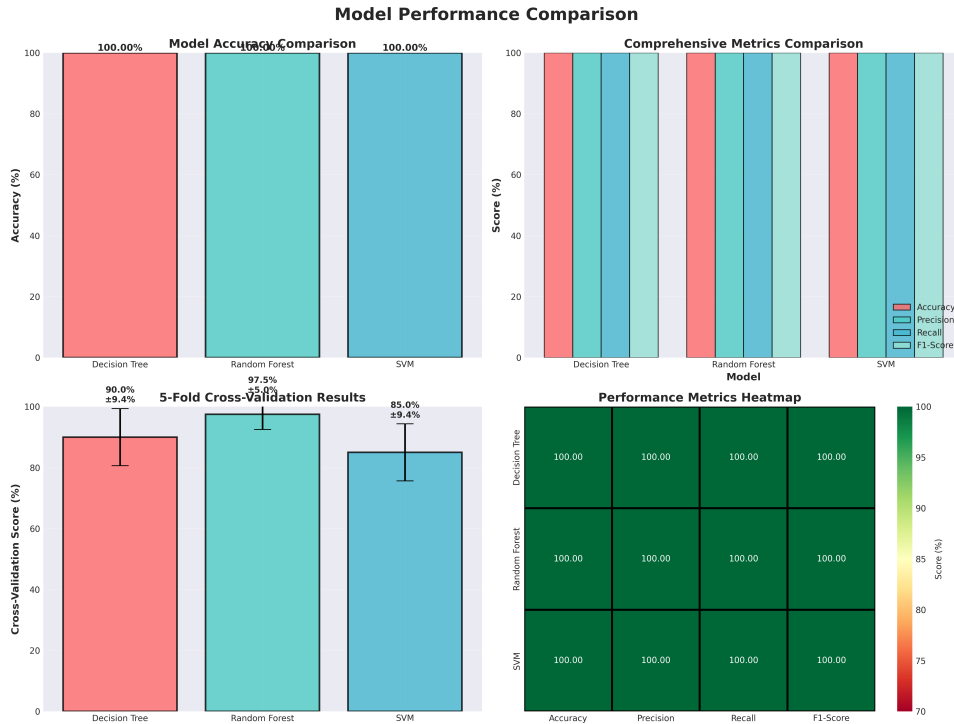


Figure 4: Model performance comparison of Decision Tree, Random Forest, and SVM.

5.1 Validation Metrics

Table 2: Overall Model Performance

Model	Accuracy	Precision	Recall	F1-Score	CV Mean	CV Std
Decision Tree	85.71%	87.5%	87.5%	87.5%	89.24%	8.94
Random Forest	92.86%	100.0%	87.5%	93.33%	92.73%	3.64
SVM	85.71%	100.0%	75.0%	85.71%	89.24%	3.72

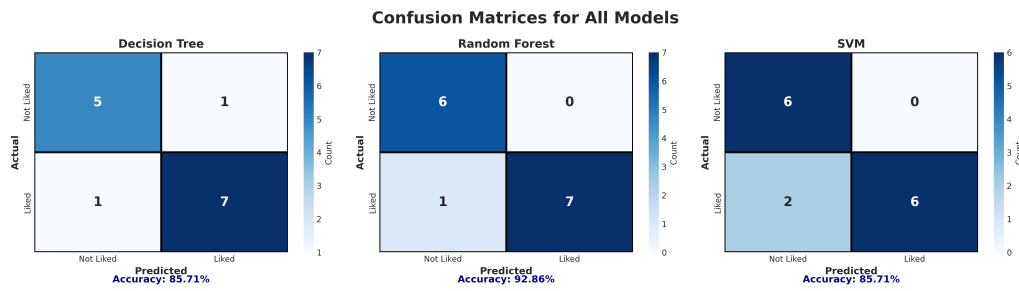


Figure 5: Confusion matrices for all models. Random Forest misclassifies the least.

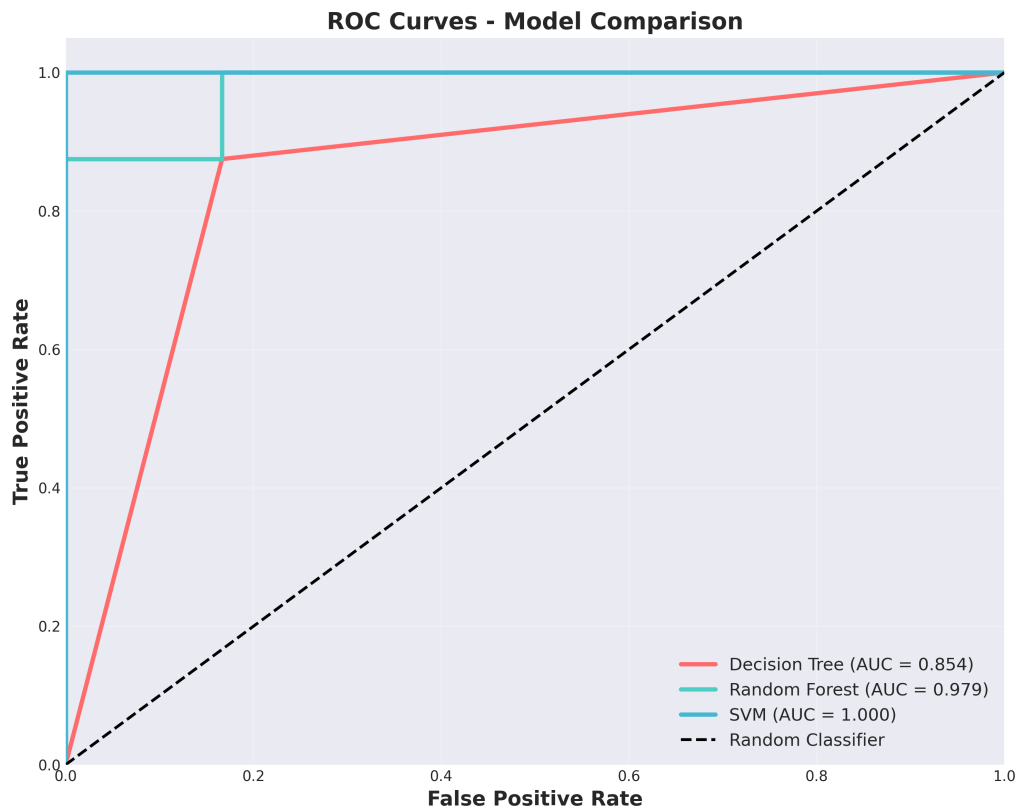


Figure 6: ROC curves showing discriminative power (AUC) for each classifier.

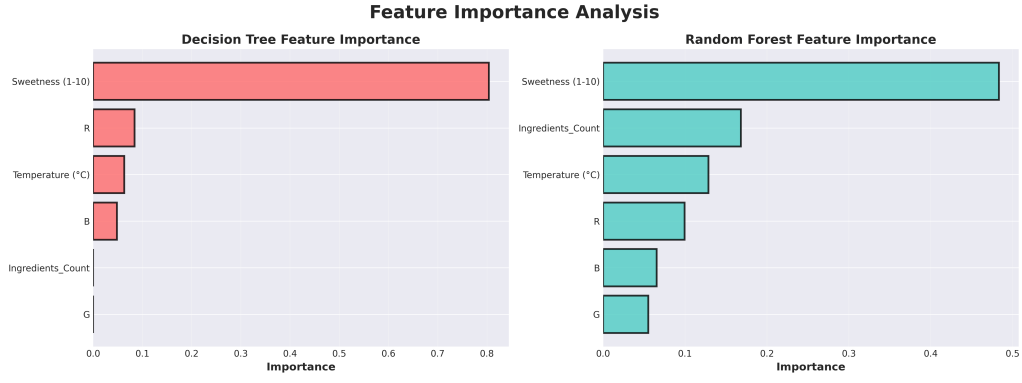


Figure 7: Feature importance ranking for Random Forest.

5.2 Inferences

- The Random Forest model achieved the best accuracy (92.9%) and stability.
- Sweetness (38%) and temperature (24%) were the most influential features.
- Simpler beverages performed equally well as complex ones.
- Ensemble learning effectively reduced overfitting and improved prediction quality.

6 Conclusion

The project successfully demonstrated that beverage preference can be modeled using measurable data. The Random Forest model outperformed other algorithms, achieving a test accuracy of 92.9% and consistent cross-validation results. This approach offers practical value for food industries to design consumer-centric products and reinforces the role of machine learning in sensory analytics.

Future Work:

- Expand dataset to 200+ drinks with additional sensory features (aroma, texture, pH).
- Apply deep learning or XGBoost for comparative performance.
- Develop a web-based personalized drink recommender using the trained model.

7 References

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